

Central Electricity Authority (Installation and Operation of Meters) Regulations, 2006

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Central Electricity Authority (Installation and Operation of Meters) Regulations, 2006

UNION OF INDIA

India

Central Electricity Authority (Installation and Operation of Meters) Regulations, 2006

Rule CENTRAL-ELECTRICITY-AUTHORITY-INSTALLATION-AND-OPERATION-OF-METERS-REGULATIONS-2006 of 2006

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Central Electricity Authority (Installation and Operation of Meters) Regulations, 2006

Published vide Notification Gazette of India, Extra; Part 3, Section 4, dated 22.3.2006.

1683.

Not. No. 502/70/CEA/DP&D, dated 17.3.2006.- In exercise of the powers conferred by sub-section (1) of section 55 and clause (e) of section 73 read with sub-section (2) of section 177 of Electricity Act, 2003, the Central Electricity Authority hereby makes the following regulations for regulating the installation and operation of meters, namely :-

1.

Short title and commencement.-

(1)

These regulations may be called the Central Electricity Authority (Installation and Operation of Meters) Regulations, 2006.

(2)

These Regulations shall come into force on the date of their publication in the Gazette of India.

2.

Definitions.-

(1)

In these regulations, unless the context other wise requires, -

(a)

`Act' means the Electricity Act, 2003;

(b)

`Accredited Test Laboratory' means a test laboratory accredited by National Accreditation Board for Testing and Calibration Laboratories (NABL);

(c)

`Active Energy' means the electricity supplied or consumed during a time interval, being the integral of Active Power with respect to time, measured in the units of `Watt-hours' or standard multiples thereof. One `kilowatt-hour' (kWh) is one unit;

(d)

`Active Power' means the electrical power, being the product of root mean square (rms) voltage, root mean square (rms) current and cosine of the phase angle between the voltage and current vectors and measured in units of `Watt' (W) or in standard multiples thereof;

(e)

`Appropriate Load Despatch Centre' means `National Load Despatch Centre' (NLDC) or `Regional Load Despatch Centre' (RLDC) or the `State Load Despatch Centre' (SLDC) as the case may be;

(f)

`Appropriate Transmission Utility' means the `Central Transmission Utility' (CTU) or the `State Transmission Utility' (STU), as the case may be;

(g)

`Availability Based Tariff (ABT)' means a tariff structure based on availability of generating units and having components, viz, Capacity Charges (CC), Energy Charges (EC) or Variable Charges (VC) and charges for Unscheduled Interchange (UI);

(h)

`Buyer' means any generating company or licensee or consumer whose system receives electricity from the system of generating company or licensee;

(i)

`Check Meter' means a meter, which shall be connected to the same core of the Current Transformer (CT) and Voltage Transformer (VT) to which main meter is connected and shall be used for accounting and billing of electricity in case of failure of main meter;

(j)

`Consumer Meter' means a meter used for accounting and billing of electricity supplied to the consumer but excluding those consumers covered under Interface Meters;

(k)

`Correct Meter' means a meter, which shall at least have, features, Accuracy Class and specifications as per the Standards on Installation and Operation of Meters given in Schedule of these Regulations;

(l)

`Energy Accounting and Audit Meters' means meters used for accounting of the electricity to various segments of electrical system so as to carry out further analysis to determine the consumption and loss of energy therein over a specified time period;

(m)

`Instrument Transformer' means the `Current Transformer' (CT), `Voltage Transformer' (VT) and `Capacitor Voltage

Transformer' (CVT);

(n)

`Interface Meter' means a meter used for accounting and billing of electricity, connected at the point of interconnection between electrical systems of generating company, licensee and consumers, directly connected to the Inter-State Transmission System or Intra-State Transmission System who have to be covered under ABT and have been permitted open access by the Appropriate Commission;

(o)

`Main Meter' means a meter, which would primarily be used for accounting and billing of electricity;

(p)

`Meter' means a device suitable for measuring, indicating and recording consumption of electricity or any other quantity related with electrical system and shall include, wherever applicable, other equipment such as Current Transformer (CT), Voltage Transformer (VT) or Capacitor Voltage Transformer (CVT) necessary for such purpose;

(q)

`Power Factor' means the cosine of the electrical angle between the voltage and current vectors in an AC electrical circuit;

(r)

`Prepaid Meter' means a meter which facilitates use of electricity only after advance payment;

(s)

`Reactive Energy' means, the integral of Reactive Power with respect to time and measured in the units of `Volt-Ampere hours reactive (VARh) or in standard multiples thereof;

(t)

`Reactive Power' means the product of root mean square (rms) voltage, root mean square (rms) current and the sine of the electrical phase angle between the voltage complexor and current complexor, measured in `Volt-ampere reactive' (VAR) and in standard multiples thereof;

(u)

`Standards' means `Standards on Installation and Operation of Meters' given in the Schedule of these Regulations unless otherwise any other standard specifically referred;

(v)

`Standby Meter' means a meter connected to CT and VT, other than those used for main meter and check meter and shall be used for accounting and billing of electricity in case of failure of both main meter and check meter;

(w)

`Supplier' means any generating company or licensee from whose system electricity flows into the system of another generating company or licensee or consumer;

(x)

`Time of the Day (TOD) Meter' means a meter suitable for recording and indicating consumption of electricity during specified time periods of the day.

(2)

The words and expressions used and not defined in these Regulations but defined in the Act shall have the meaning assigned to them in the Act.

3.

Applicability of regulations.-

(1)

These Regulations shall be applicable to meters installed and to be installed by all the generating companies and licensees who are engaged in the business of generation, transmission, trading, distribution, supply of electricity and to all categories of consumers.

(2)

After coming into force of these regulations, the provisions of the Indian Electricity Rules, 1956 relating to installation and operation of meters in this regard shall not be applicable.

(3)

These regulations provide for type, standards, ownership, location, accuracy class, installation, operation, testing and maintenance, access, sealing, safety, meter reading and recording, meter failure or discrepancies, anti tampering

features, quality assurance, calibration and periodical testing of meters, additional meters and adoption of new technologies in respect of following meters for correct accounting, billing and audit of electricity:

(i)

Interface meter

(ii)

Consumer meter

(iii)

Energy accounting and audit Meter

4.

Type of meters.-

(1)

All interface meters, consumer meters and energy accounting and audit meters shall be of static type.

(2)

The meters not complying with these regulations shall be replaced by the licensee on his own or on request of the consumer. The meters may also be replaced as per the regulations or directions of the Appropriate Commission or pursuant to the reforms programme of the Appropriate Government.

5.

Standards.-

All interface meters, consumer meters and energy accounting and audit meters shall -

(a)

comply with the relevant standards of Bureau of Indian Standards (BIS). If BIS Standards are not available for a particular equipment or material, the relevant British Standards (BS), International Electro-technical Commission (IEC) Standards, or any other equivalent Standard shall be followed:

Provided that whenever an international Standard or IEC Standard is followed, necessary corrections or modifications shall be made for nominal system frequency, nominal system voltage, ambient temperature, humidity and other conditions prevailing in India before actual adoption of the said Standard;

(b)

conform to the standards on 'Installation and Operation of Meters' as specified in Schedule annexed to these regulations and as amended from time to time.

6.

Ownership of meters.-

(1)

Interface meters.-

(a)

All interface meters installed at the points of interconnection with Inter-State Transmission System (ISTS) for the purpose of electricity accounting and billing shall be owned by CTU.

(b)

All interface meters installed at the points of interconnection with Intra-State Transmission System excluding the system covered under sub-clause (a) above for the purpose of electricity accounting and billing shall be owned by STU.

(c)

All interface meters installed at the points of inter connection between the two licensees excluding those covered under sub-clauses (a) and (b) above for the purpose of electricity accounting and billing shall be owned by respective licensee of each end.

(d)

All interface meters installed at the points of inter connection for the purpose of electricity accounting and billing not covered under sub-clauses (a), (b) and (c) above shall be owned by supplier of electricity.

(2)

Consumer meters.-

(a)

Consumer meters shall generally be owned by the licensee.

(b)

If any consumer elects to purchase a meter, the same may be purchased by him. Meter purchased by the consumer shall be tested, installed and sealed by the licensee. The consumer shall claim the meter purchased by him as his asset only after it is permanently removed from the system of the licensee.

(c)

All consumer meters shall bear BIS mark, meet the requirements of these regulations and have additional features as approved by the Appropriate Commission or pursuant to the reforms programme of the Appropriate Government. To facilitate this, the licensee shall provide a list of makes and models of the meters.

(3)

Energy accounting and audit meters.- Energy accounting and audit meters shall be owned by the generating company or licensee, as the case may be.

7.

Locations of meters.-

(1)

The location of interface meters, consumer meters and energy accounting and audit meters shall be as per the Table given below :

Provided that the generating companies or licensees may install meters at additional locations in their systems depending upon the requirement.

TABLE

Sl. No.

Stages

Main meter

Check meter

Stand by meter

A.

Generating Station

On all outgoing feeders.

On all outgoing feeders.

(i) High Voltage (HV) side of Generator Transformers

(ii) High Voltage(HV) side of all Station Auxiliary Transformers

Explanation:- The location of main, check and standby meters installed at the existing generating stations shall not be changed unless permitted by the Authority.

Sl.

No.

Stages

Main Meter

Check Meter

Stand by Meter

B.

Transmission and Distribution System

At one end of the line between the sub-stations of the same licensee, and at both ends of the line between sub-stations of two different licensees. Meters at both ends shall be considered as main meters for respective licensees

-

There shall be no separate standby meter. Meter installed at other end of the line in case of two different licensees shall work as standby meter.

C.

Inter-Connecting Transformer (ICT)

High Voltage (HV) side of ICT.

-

Low Voltage (LV) side of ICT.

D.

Consumer directly connected to the Inter-State Transmission System or Intra-State Transmission System who have to

be covered under ABT and have been permitted open access by the Appropriate Commission or Any other system not covered above

As decided by the Appropriate Commission.

(a)

Interface Meters.-

(i)

Consumers who have interconnection with the Inter-State Transmission System or Intra-State Transmission System and have been permitted open access by the Appropriate Commission shall be provided with interface meters.

(ii)

For consumers connected to distribution system and permitted open access, provision of interface meters may be made as per the regulations or directions of the Appropriate Commission.

(iii)

The scheme for location of interface meters shall be submitted to the CTU or the STU or the CTU or the STU or the licensee by owner of the meter in advance, before the installation of the scheme.

(b)

Consumer meters.-

(i)

The consumer meter shall be installed by the licensee either at consumer premises or outside the consumer premises: Provided that where the licensee installs the meter outside the premises of the consumer, then the licensee shall provide real time display unit at the consumer premises for his information to indicate the electricity consumed by the consumer:

Provided further that for the billing purpose, reading of consumer meter and not the display unit shall be taken into account.

(ii)

In the event the Appropriate Commission allows supply of electricity directly from a generating company to consumer on a dedicated transmission system, the location of the meter will be as per their mutual agreement.

(c)

Energy accounting and audit meters.- Energy accounting and audit meters shall be installed at such locations so as to facilitate to account for the energy generated, transmitted, distributed and consumed in the various segments of the power system and the energy loss. The location of these meters shall be as under:

(i)

Generating Stations

(1)

at the stator terminal of the generator;

(2)

on HV and LV sides of the station and the unit auxiliary transformers;

(3)

on feeders to various auxiliaries.

(ii)

Transmission System

All incoming and out going feeders (If the interface meters do not exist).

(iii)

Distribution System

(1)

all incoming feeders (11 kV and above);

(2)

all outgoing feeders (11 kV and above);

(3)

Sub-Station Transformer including Distribution Transformer - Licensee may provide the meter on primary or secondary side or both sides depending upon the requirement for energy accounting and audit.

8.

Accuracy Class of meters.-

Every meter shall meet the requirement of accuracy class as specified in the standards given in the Schedule.

9.

Installation of meters.-

(1)

Generating company or licensee, as the case may be, shall examine, test and regulate all meters before installation and only correct meters shall be installed.

(2)

The meter shall be installed at locations, which are easily accessible for installation, testing, commissioning, reading, recording and maintenance. The place of installation of meter shall be such that minimum inconvenience and disruptions are caused to the site owners and the concerned organizations.

(3)

In case of single phase meters, the consumer shall ensure that there is no common neutral or phase or looping of neutral or phase of two or more consumers on consumers' side wiring. If such common neutral or phase or looping of neutral or phase comes to the notice of the licensee, it shall suitably inform the consumer through installation report or regular electricity bills or meter test report as applicable.

(4)

Consumer shall install the Earth Leakage Protective Device (ELPD) in accordance with the provisions of the rules or regulations in this regard.

(5)

If the earth leakage indication is displayed in the meter the licensees shall suitably inform the consumer through installation report or regular electricity bills or meter test report as applicable.

(6)

In case CTs and VTs form part of the meters, the meter shall be installed as near the instrument transformers as possible to reduce the potential drop in the secondary leads.

10.

Operation, Testing and Maintenance of meters.-

The operation, testing and maintenance of all types of meters shall be carried out by the generating company or the licensee, as the case may be.

11.

Access to meter.-

The owner of the premises where, the meter is installed shall provide access to the authorized representative(s) of the licensee for installation, testing, commissioning, reading and recording and maintenance of meters.

12.

Sealing of meters.-

(1)

Sealing Arrangements.-

(a)

All meters shall be sealed by the manufacturer at its works. In addition to the seal provided by the manufacturer at its works, the sealing of all meters shall be done as follows at various sealing points as per the standards given in the Schedule:

(i)

Sealing of interface meters, shall also be done by both the supplier and the buyer.

(ii)

Sealing of consumer meters shall be done by the licensee.

(iii)

Sealing of energy accounting and audit meters shall be done by the licensee or generating company as the case may be.

(b)

A tracking and recording software for all new seals shall be provided by the manufacturer of the meter so as to track total movement of seals starting from manufacturing, procurement, storage, record keeping, installation, series of

inspections, removal and disposal.

(c)

Seal shall be unique for each utility and name or logo of the utility shall be clearly visible on the seals.

(d)

Only the patented seals (seal from the manufacturer who has official right to manufacture the seal) shall be used.

(e)

Polycarbonate or acrylic seals or plastic seals or holographic seals or any other superior seal shall be used.

(f)

Lead seals shall not be used in the new meters. Old lead seals shall be replaced by new seals in a phased manner and the time frame of the same shall be submitted by the licensee to the Appropriate Commission for approval.

(2)

Removal of seals from meters.-

(a)

Interface meters.- Whenever seals of the interface meters have to be removed for any reason, advance notice shall be given to other party for witnessing the removal of seals and resealing of the interface meter. The breaking and re-sealing of the meters shall be recorded by the party, who carried out the work, in the meter register, mentioning the date of removal and resealing, serial numbers of the broken and new seals and the reason for removal of seals.

(b)

Consumer meters.- Seal of the consumer meter shall be removed only by the licensee. No consumer shall tamper with, break or remove the seal under any circumstances. Any tampering, breaking or removing the seal from the meter shall be dealt with as per relevant provisions of the Act.

(c)

Energy accounting and audit meters.- Seal of the energy accounting and audit meter shall be removed only by the generating company or the licensee who owns the meter.

13.

Safety of meters.-

(1)

The supplier or buyer in whose premises the interface meters are installed shall be responsible for their safety.

(2)

The consumer shall, as far as circumstances permit, take precautions for the safety of the consumer meter installed in his premises belonging to the licensee.

(3)

Licensee shall be responsible for the safety of the consumer meter located outside the premises of the consumer and the consumer shall be responsible for the safety of the real time display unit installed by the licensee in consumer premises.

(4)

The generating company or the licensee who owns the energy accounting and audit meters shall be responsible for its safety.

14.

Meter reading and recording.-

(1)

Interface meters.- It shall be the responsibility of the Appropriate Transmission Utility or the licensee to take down the meter reading and record the metered data, maintain database of all the information associated with the interface meters and verify the correctness of metered data and furnish the same to various agencies as per the procedure laid down by the Appropriate Commission.

(2)

Consumer meters.-

(a)

It shall be the responsibility of the licensee to record the metered data, maintain database of all the information associated with the consumer meters and verify the correctness of metered data.

(b)

The licensee shall maintain accounts for the electricity consumption and other electrical quantities of its consumers.

(c)

Brief history, date of installation and details of testing, calibration and replacement of meters shall be maintained by the licensee.

(3)

Energy accounting and audit meters.- It shall be the responsibility of the generating company or licensee to record the metered data, maintain database of all the information associated with the energy accounting and audit meters and verify the correctness of metered data. Each generating company or licensee shall prepare quarterly, half-yearly and yearly energy account for its system for taking appropriate action for efficient operation and system development.

15.

Meter failure or discrepancies.-

(1)

Interface meters.-

(a)

Whenever difference between the readings of the Main meter and the Check meter for any month is more than 0.5%, the following steps shall be taken:

(i)

checking of CT and VT connections;

(ii)

testing of accuracy of interface meter at site with reference standard meter of accuracy class higher than the meter under test.

If the difference exists even after such checking or testing, then the defective meter shall be replaced with a correct meter.

(b)

In case of conspicuous failures like burning of meter and erratic display of metered parameters and when the error found in testing of meter is beyond the permissible limit of error provided in the relevant standard, the meter shall be immediately replaced with a correct meter.

(c)

In case where both the Main meter and Check meter fail, at least one of the meters shall be immediately replaced by a correct meter.

(d)

Billing for the Failure period:

(i)

The billing for the failure period of the meter shall be done as per the procedure laid down by the Appropriate Commission.

(ii)

Readings recorded by Main, Check and Standby meters for every time slot shall be analysed, crosschecked and validated by the Appropriate Load Despatch Centre (LDC). The discrepancies, if any, noticed in the readings shall be informed by the LDC in writing to the energy accounting agency for proper accounting of energy. LDC shall also intimate the discrepancies to the Appropriate Transmission Utility or the licensee, who shall take further necessary action regarding testing, calibration or replacement of the faulty meters in accordance with the provisions laid down.

(e)

The defective meter shall be immediately tested and calibrated

(2)

Consumer meters.- In case the consumer reports to the licensee about consumer meter readings not commensurate with his consumption of electricity, stoppage of meter, damage to the seal, burning or damage of the meter, the licensee shall take necessary steps as per the procedures given in the Electricity Supply Code of the Appropriate Commission read with the notified conditions of supply of electricity.

(3)

Energy accounting and audit meters.- Energy accounting and audit meters shall be rectified or replaced by the

generating company or licensee immediately after notice of any of the following abnormalities:

- (a)
the errors in the meter readings are outside the limits prescribed for the specified Accuracy Class;
- (b)
meter readings are not in accordance with the normal pattern of the load demand;
- (c)
meter tampering, or erratic display or damage.

16.

Anti-tampering features of meters.-

The meters shall be provided with such anti-tampering features as per the Standards on Installation and Operation of Meters given in the Schedule.

17.

Quality assurance of meters.-

(1)

The distribution licensee shall put in place a system of quality assurance and testing of meters with the approval of Appropriate Commission.

(2)

The licensee shall set up appropriate number of accredited testing laboratories or utilize the services of other accredited testing laboratories. The licensee shall take immediate action to get the accreditations of their existing meter testing laboratories from NABL, if not already done.

(3)

The generating company or licensee shall ensure that all type, routine and acceptance tests are carried out by the manufacturer complying with the requirement of the relevant IS or BS or IEC as the case may be.

18.

Calibration and periodical testing of meters.-

(1)

Interface meter

(a)

At the time of commissioning, each interface meter shall be tested by the owner at site for accuracy using standard reference meter of better accuracy class than the meter under test.

(b)

All interface meters shall be tested at least once in five years. These meters shall also be tested whenever the energy and other quantities recorded by the meter are abnormal or inconsistent with electrically adjacent meters. Whenever there is unreasonable difference between the quantity recorded by interface meter and the corresponding value monitored at the billing center via communication network, the communication system and terminal equipment shall be tested and rectified. The meters may be tested using NABL accredited mobile laboratory or at any accredited laboratory and recalibrated if required at manufacturer's works.

(c)

Testing and calibration of interface meters may be carried out in the presence of the representatives of the supplier and buyer. The owner of the meter shall send advance notice to the other party regarding the date of testing.

(2)

Consumer meters.- The testing of consumer meters shall be done at site at least once in five years. The licensee may instead of testing the meter at site can remove the meter and replace the same by a tested meter duly tested in an accredited test laboratory. In addition, meters installed in the circuit shall be tested if study of consumption pattern changes drastically from the similar months or season of the previous years or if there is consumer's complaint pertaining to a meter. The standard reference meter of better accuracy class than the meter under test shall be used for site testing of consumer meters up to 650 volts. The testing for consumers meters above 650 volts should cover the entire metering system including CTs, VTs. Testing may be carried out through NABL accredited mobile laboratory using secondary injection kit, measuring unit and phantom loading or at any accredited test laboratory and recalibrated if required at manufacturer's works.

(3)

Energy accounting and audit meters.- Energy accounting and audit meters shall be tested at site at least once in five years or whenever the accuracy is suspected or whenever the readings are inconsistent with the readings of other meters, e.g., check meters, standby meters. The testing must be carried out without removing the CTs and VTs connection. Testing may be carried out through NABL accredited mobile laboratory using secondary injection kit, measuring unit and phantom loading or at any accredited test laboratory and recalibrated if required at manufacturer's works.

19.

Additional meters.-

In addition to any meter which may be placed for recording the electricity consumed by the consumer, the licensee may connect additional meters, maximum demand indicator or other apparatus as he may think fit for the purposes of ascertaining or regulating either the quantity of electricity supplied to the consumer, or the number of hours during which the supply is given, or the rate per unit of time at which energy is supplied to the consumer, or any other quantity or time connected with the supply to consumer:

Provided that the meter, indicator or apparatus shall not, in the absence of an agreement to the contrary, be placed otherwise than between the distributing mains of the licensee and any meter:

Provided further that, where the charges for the supply of energy depend wholly or partly upon the reading or indication of any such meter, indicator or apparatus as aforesaid, the licensee shall, in the absence of an agreement to the contrary, keep the meter, indicator or apparatus correct.

20.

Adoption of new technologies.-

The distribution licensee shall make out a plan for introduction and adoption of new technologies such as pre-paid meters, time of the day meters (TOD), automatic remote meter reading system through appropriate communication system with the approval of the Appropriate Commission or as per the regulations or directions of the Appropriate Commission or pursuant to the reforms programme of the Appropriate Government.

Schedule

(see regulations 2, 5, 8, 12 and 16)

Standards on Installation and Operation of Meters

Part I ? Standards Common To All Type of Meters

(1)

These standards provide for specification of meters, immunity to external factors, sealing points and functional requirements that are required from regulatory perspective. Detailed technical specification shall be prepared by the purchaser of the meter.

(2)

Specifications of meters.-

Standard Reference Voltage

As per IS

Voltage Range

As per IS

Standard Frequency

As per IS

Standard Basic Current

As per IS (Current range of consumer meters shall be so chosen as to record the load current corresponding to the sanctioned load)

Accuracy Class

Meters shall meet the following requirements of Accuracy Class:

{|

Interface meters

0.2S

Consumer meters

Up to 650 volts

1.0 or better

Above 650 volts and up to 33 kilo volts

0.5S or better

Above 33 kilo volts

0.2S

Energy accounting and audit meters: - The accuracy class of meters in generation and transmission system shall not be inferior to that of 0.2S Accuracy Class. The accuracy class of meters in distribution system shall not be inferior to that of 0.5S Accuracy Class

| -

| Starting Current and Maximum

| -

| -

| Current

| As per IS

| -

| Power Factor Range

| As per IS

| -

| Power Frequency Withstand

| -

| -

| Voltage

| As per IS

| -

| Impulse Voltage Withstand

| -

| -

| Test for 1.2/50 micro sec

| As per IS

| -

| Power Consumption

| As per IS

| }

(3)

Meter shall have downloading facilities of metered data through Meter Reading Instrument (MRI).

(4)

Immunity to External Factors.- The meter shall be immune to external influences like magnetic induction, vibration, electrostatic discharge, switching transients, surge voltages, oblique suspension and harmonics and necessary tests shall be carried out in accordance with relevant standard.

(5)

Sealing Points.- Sealing shall be done at the following points (as applicable):

(a)

Meter body or cover

(b)

Meter terminal cover

(c)

Meter test terminal block

(d)

Meter cabinet

(6)

The accuracy class of Current transformers (CTs) and Voltage transformers (VTs) shall not be inferior to that of associated meters. The existing CTs and VTs not complying with these regulations shall be replaced by new CTs and

VTs, if found defective, non-functional or as per the directions of the Appropriate Commission. In case the CTs and VTs of the same Accuracy Class as that of meters can not be accommodated in the metering cubicle or panel due to space constraints, the CTs and VTs of the next lower Accuracy Class can be installed.

(7)

The Voltage Transformers shall be electromagnetic VT or Capacitive Voltage Transformer (CVT).

Part II ? Standards for interface meters

(1)

Functional Requirements.-

(a)

The Interface meters suitable for ABT shall be static type, composite meters, as self-contained devices for measurement of active and reactive energy, and certain other parameters as described in the following paragraphs. The meters shall be suitable for being connected directly to voltage transformers (VTs) having a rated secondary line-to-line voltage of 110 V, and to current transformers (CTs) having a rated secondary current of 1A (Model-A :3 element 4 wire or Model C: 2 element, 3 wire) or 5A (model-B: 3 element, 4 wire or Model D: 2 element 3 wire). The reference frequency shall be 50Hz.

(b)

The meters shall have a non-volatile memory in which the following shall be automatically stored:

(i)

Average frequency for each successive 15-minute block, as a two digit code (00 to 99 for frequency from 49.0 to 51.0Hz).

(ii)

Net Wh transmittal during each successive 15-minute block, upto second decimal, with plus/minus sign.

(iii)

Cumulative Wh transmittal at each midnight, in six digits including one decimal.

(iv)

Cumulative VARh transmittal for voltage high condition, at each midnight, in six digits including one decimal.

(v)

Cumulative VARh transmittal for voltage low condition, at each midnight, in six digits including one decimal.

(vi)

Date and time blocks of failure of VT supply on any phase, as a star(*) mark.

(c)

The meters shall store all the above listed data in their memories for a period of at least ten days. The data older than ten days shall get erased automatically. Each meter shall have an optical port on its front for tapping all data stored in its memory using a hand held data collection device. The meter shall be suitable for transmitting the data to remote location using appropriate communication medium.

(d)

The active energy (Wh) measurement shall be carried out on 3-phase, 4-wire principle, with an accuracy as per class 0.2 S of IEC-687/IEC-62053-22. In model A and C, the energy shall be computed directly in CT and VT secondary quantities, and indicated in watt-hours. In model-B and Model D, the energy display and recording shall be one fifth of the Wh computed in CT and VT secondary quantities.

(e)

The Var and reactive energy measurement shall also be on 3-phase, 4-wire principle, with an accuracy as per class 2 of IEC-62053-23 or better. In model-A or Model C, the Var and VARh computation shall be directly in CT and VT secondary quantities. In model-B or Model D, the above quantities shall be displayed and recorded as one-fifth of those computed in CT and VT secondary quantities. There shall be two reactive energy registers, one for the period when average RMS voltage is above 103% and the other for the period the voltage is below 97%.

(f)

The 15-minute Wh shall have a +ve sign when there is a net Wh export from substation busbars, and a -ve sign when there is a net Wh import. The integrating (cumulative) registers for Wh and Varh shall move forward when there is Wh/Varh export from substation busbars, and backward when there is an import.

(g)

The meters shall also display (on demand), by turn, the following parameters :

- (i)
Unique identification number of the meter
- (ii)
Date
- (iii)
Time
- (iv)
Cumulative Wh register reading
- (v)
Average frequency of the previous 15-minute block
- (vi)
Net Wh transmittal in the previous 15-minute block, with +/-sign
- (vii)
Average percentage voltage
- (viii)
Reactive power with +/- sign
- (ix)
Voltage-high VARh register reading
- (x)
Voltage-low VARh register reading.
- (h)

The three line-to-neutral voltages shall be continuously monitored, and in case any of these falls below 70%, the condition shall be suitably indicated and recorded. The meters shall operate with the power drawn from the VT secondary circuits, without the need for any auxiliary power supply. Each meter shall have a built-in calendar and clock, having an accuracy of 30 seconds per month or better.

- (i)
The meters shall be totally sealed and tamper-proof, with no possibility of any adjustment at site, except for a restricted clock correction. The harmonics shall be filtered out while measuring Wh, Var and VARh, and only fundamental frequency quantities shall be measured/computed.

- (j)
The Main meter and the Check meter shall be connected to same core of CTs and VTs.

Part III ? Standards for consumers meters

- (1)
Measuring Parameters:-

- (a)
The consumer meter shall be suitable for measurement of cumulative active energy utilized by the consumer.

- (b)
The consumer meter may have the facilities to measure, record and display one or more of the following parameters depending upon the tariff requirement for various categories of consumers. All parameters excluding instantaneous electrical parameters shall also be stored in memory.

- (i)
Cumulative reactive energy
- (ii)
Average power factor
- (iii)
Time of use of energy
- (iv)
Apparent power
- (v)
Maximum demand

(vi)

Phase voltage and line currents

(2)

All the three phase meters shall have data storage capacity for at least 35 days in a non-volatile memory.

(3)

Anti-Tampering Features.-

(a)

The meter shall not get damaged or rendered non-functional even if any phase and neutral are interchanged.

(b)

The meter shall register energy even when the return path of the load current is not terminated back at the meter and in such a case the circuit shall be completed through the earth. In case of metallic bodies, the earth terminal shall be brought out and provided on the outside of the case.

(c)

The meter shall work correctly irrespective of the phase sequence of supply (only for poly phase).

(d)

In the case of 3 phase, 3 wire meter even if reference Y phase is removed, the meter shall continue to work. In the case of 3 phase, 4 wire system, the meter shall keep working even in the presence of any two wires i.e., even in the absence of neutral and any one phase or any two phases.

(e)

In case of whole current meters and LV CT operated meter, the meter shall be capable of recording energy correctly even if input and output terminals are interchanged.

(f)

The registration must occur whether input phase or neutral wires are connected properly or they are interchanged at the input terminals.

(g)

The meter shall be factory calibrated and shall be sealed suitably before dispatch.

(h)

The meter shall be capable of recording occurrences of a missing potential (only for VT operated meters) and its restoration with date and time of first such occurrence and last restoration along with total number and duration of such occurrences during the above period for all phases.

(i)

Additional anti-tampering features including logging of tampers such as current circuit reversal, current circuit short or open and presence of abnormal magnetic field may be provided as per the regulations or directions of the Appropriate Commission or pursuant to the reforms programme of the Appropriate Government.

Part IV ? Standards for energy accounting and audit meters

(1)

The energy accounting and audit meters shall be suitable for measurement, recording and display of cumulative active energy with date and time.

(2)

The energy accounting and audit meters may also have the facility to measure, record and display one or more of the following parameters depending upon the energy accounting and audit requirement. All parameters excluding instantaneous electrical parameters shall also be stored in memory.

(a)

Apparent power

(b)

Phase wise kilowatt at peak KVA

(c)

Phase wise KVA(reactive) at peak KVA

(d)

Phase wise voltage at peak KVA

(e)

Power down time

(f)

Average power factor

(g)

Line currents

(h)

Phase voltages

(i)

Date and time

(j)

Tamper events

(3)

The energy accounting and audit meter shall have data storage capacity for at least 35 days in a non-volatile memory.

(4)

Energy accounting and audit meters shall have facility to download the parameters through meter reading instruments as well as remote transmission of data over communication network.

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